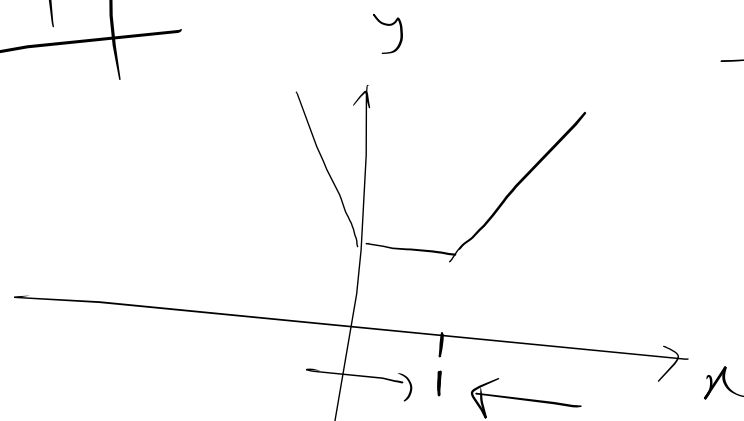


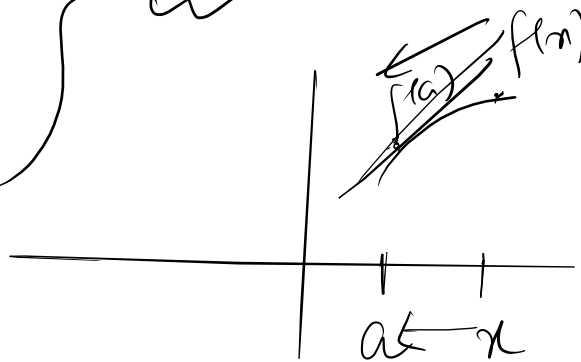
$$f(x) = |x| + |x-1|, \quad \boxed{x=0, 1}$$

$$f(x) = \begin{cases} -2x+1, & x < 0 \\ 1, & 0 \leq x < 1 \\ 2x-1, & x \geq 1 \end{cases}$$



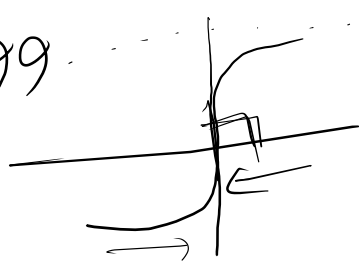
$$L.H.D. = \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h}$$

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$



$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{a+h-a}$$

$$\frac{0.9999 \dots}{99.9999 \dots}$$

$$2. \quad f(x) = \begin{cases} \underline{x^2 - 16x}, & \underline{x < 9} \quad 8.999 \dots \\ 12\sqrt{x}, & \underline{x \geq 9} \quad 9.00000 \dots \end{cases}$$


Differentiable $\Rightarrow$ Continuous $\Rightarrow$ limit exists

$\Leftarrow$ $\Leftarrow$

$|x|$ is continuous at $x=0$ but not differentiable at $x=0$

(i) $f(9) = 12\sqrt{9} = 12 \cdot 3 = 36$

(ii) L.H.L. $= \lim_{x \rightarrow 9^-} x^2 - 16x = 81 - 16 \times 9 = -63$

R.H.L. $= \lim_{x \rightarrow 9^+} 12\sqrt{x} = 36$

$\lim_{x \rightarrow 9} f(x)$ does not exist

$f(x)$ is not continuous at $x=9$. $f(9)=36$.

$$\begin{aligned} \text{L.H.D.} &= \lim_{h \rightarrow 0^-} \frac{f(9+h) - f(9)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{h^2 + 2h - 63 - 36}{h} \end{aligned}$$

$$= \lim_{h \rightarrow 0^-} \frac{h^2 + 2h - 99}{h} = -\infty$$

$$\begin{aligned} f(9+h) &= (9+h)^2 - 16(9+h) \\ &= 81 + 18h + h^2 - 144 - 16h \\ &= h^2 + 2h - 63 \end{aligned}$$

$$f(9+h) = 12\sqrt{9+h}$$

$$\begin{aligned} \text{R.H.D.} &= \lim_{h \rightarrow 0^+} \frac{f(9+h) - f(9)}{h} = \lim_{h \rightarrow 0^+} \frac{12\sqrt{9+h} - 36}{h} \end{aligned}$$

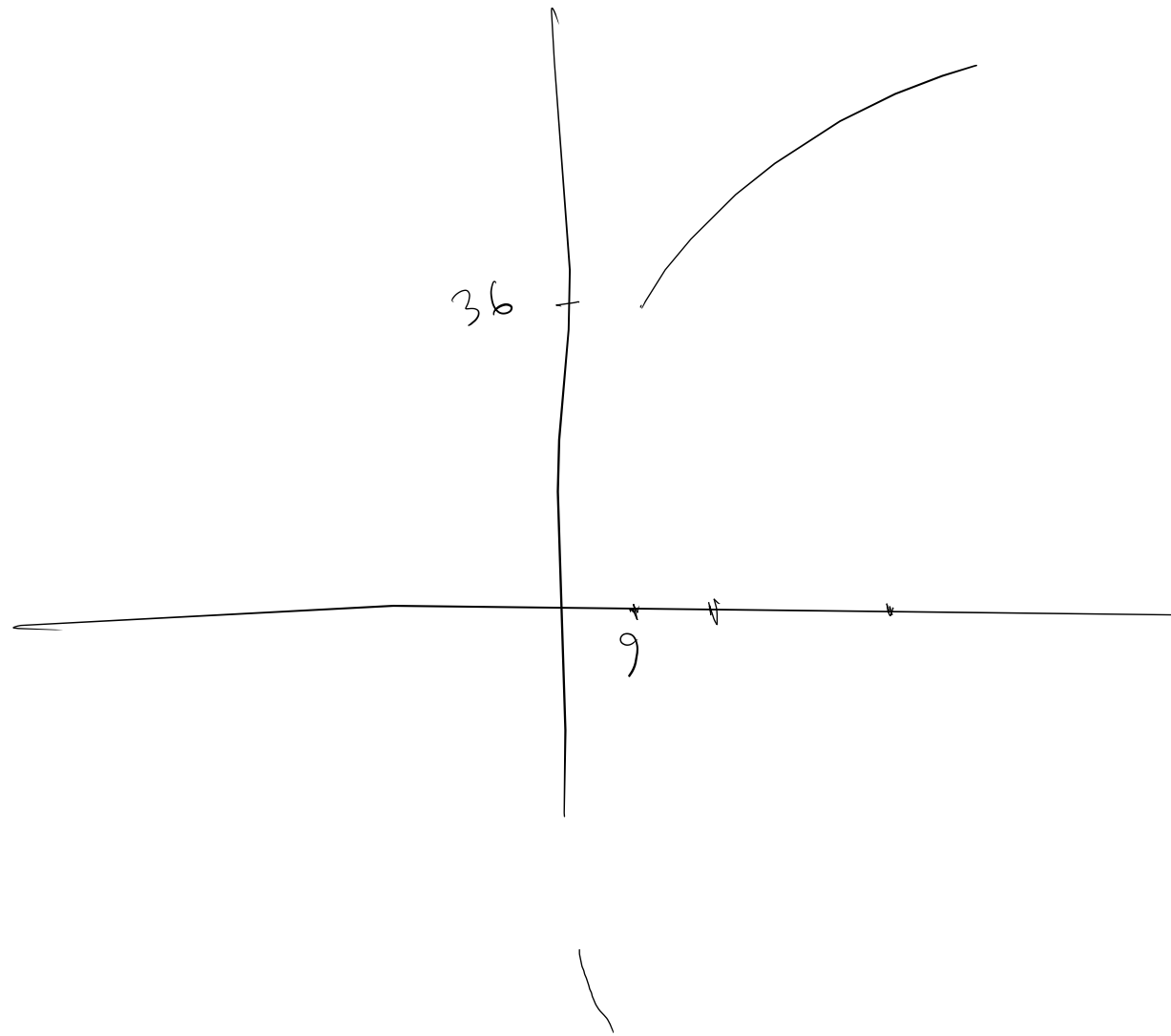
$$= \lim_{h \rightarrow 0^+} \frac{12(\sqrt{9+h} - 3)}{h} = \lim_{h \rightarrow 0^+} \frac{12(\sqrt{9+h} - 3)(\sqrt{9+h} + 3)}{h(\sqrt{9+h} + 3)} = \frac{1}{6} \quad (\text{Verify})$$

$f(x)$ is not differentiable at $x=9$. $(y-h) = (\underbrace{x-k})^2$

$$y = \underbrace{x^2 - 16x + 64} - 64 = (x-8)^2 - 64 //$$

$$y = 125x$$

$$\lim_{h \rightarrow 100} \frac{\Delta y}{\Delta x}$$



8.99999

$$1. \quad x = \sin \sqrt{y}$$

$$\Rightarrow \sqrt{y} = \sin^{-1} x$$

$$\Rightarrow \underline{y} = (\sin^{-1} x)^2$$

$$\Rightarrow y_1 = 2 \cdot \frac{\sin^{-1} x}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1 (\sqrt{1-x^2}) = 2 \sin^{-1} x$$

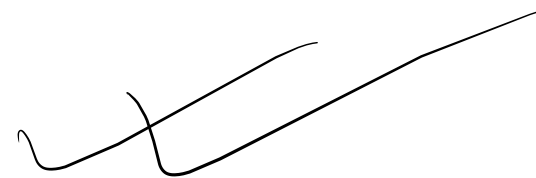
$$\Rightarrow (1-x^2) y_1^{\checkmark} = 2 (\sin^{-1} x)^2$$

$$\Rightarrow \underbrace{(1-x^2)}_u \underbrace{y_1^{\checkmark}}_v = 4y$$

$$(1-x^2) \underline{y_2} - x y_1 = 2 \quad \checkmark \checkmark$$

$$\Rightarrow (1-x^2) \cdot \underline{2y_1 y_2} + \underline{y_1^{\checkmark}} (-2x) = \underline{4y_1}$$

$$\Rightarrow (1-x^2) y_2 - x y_1 = 2.$$



$$8. \quad y = e^{\underline{m \sin^2 x}}$$

$$(1-x^2) \underline{y_2} - x y_1 - \underline{m^2 y} = 0.$$

$$\Rightarrow y_1 = m \frac{e^{\underline{m \sin^2 x}}}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1 (\sqrt{1-x^2}) = m y$$

$$\Rightarrow y_1' (1-x^2) = m^2 y$$

$$\Rightarrow (1-x^2) 2 y_1 y_2 + y_1' (-2x) = m^2 \cdot 2 y y_1$$

$$\Rightarrow (1-x^2) y_2 - x y_1 - m^2 y = 0.$$

$$10. \tan(\ln y) = x$$

$$\Rightarrow \ln y = \tan^{-1} x$$

$$\Rightarrow e^{\ln y} = e^{\tan^{-1} x}$$

$$\Rightarrow y = e^{\tan^{-1} x}$$

$$\Rightarrow y' = \frac{e^{\tan^{-1} x}}{1+x^2}$$

H.W.

$$11. \ln y = \tan^{-1} x$$

$$\Rightarrow e^{\ln y} = e^{\tan^{-1} x}$$

$$\Rightarrow y = e$$

